

Deep Research Agent - 21-Agent Upgrade

Production-Ready Enhancement Report

Current Agent Framework (21 Agents)

A. Core Pipeline Agents (7)

1. Query Decomposer: Breaks complex queries into optimal sub-questions using phi3:mini; conserves LLM budget; generates 8-12 sub-questions
2. Research Orchestrator: Coordinates multi-strategy search across arXiv/DuckDuckGo/Wikipedia/PixelRAG; manages search budget and iterations; tracks search progress
3. Multi-Source Scraper: 4-tier fallback: Trafilatura -> BS4 -> Jina Reader -> Playwright; SQLite page cache (scraper_cache.db); anti-blocking rotation
4. Domain Intelligence: Live domain credibility scoring; maintains dynamic blacklist; updates SOURCE_DOWNWEIGHT_DOMAINS automatically; credibility tiering
5. Citation Verifier: Cross-checks numeric claims + entity verification via _cross_check_numeric_claims + _verify_entity_names; flags unverified claims for re-search
6. Quality Scorer: Computes 5-dimension scores (relevance, depth, novelty, coherence, citation_accuracy) via real_quality_scorer.compute_quality_scores();
7. Coherence Auditor: Analyzes heading hierarchy (H1/H2/H3 count); transition density; list item ratio; summary/ref presence; assigns coherence score 0-10;

B. Enhancement Agents (8)

8. Lesson Learner: Extracts lessons from evaluator output; stores to Supabase knowledge_base + global KG; feeds back to planner_node via prior_lessons; enables adaptive search
9. Gap Analyzer: Identifies missing topics by comparing covered vs. missing topics; generates new sub-questions based on gap type; controls gap_detector logic
10. Repetition Detector: Detects near-duplicate sections via _cross_section_duplication_check (similarity >= 0.3); triggers source diversification when redundancy detected
11. Tone & Style Adjuster: Checks brand voice compliance; adjusts formality level (academic/industry/general); optimizes for target audience; ensures consistency
12. Fact Checker: Verifies all factual claims in report; generates "Evidence Verification Notes" section listing unverified claims; integrates with _cross_check_numeric_claims
13. Source Diversifier: When redundancy detected (repetition_detector), searches alternative domains and sources; prevents same content from different URLs
14. Export Specialist: Generates final output in quality-tiered formats: PDF (formatted), Markdown (raw data), JSON (structured); quality levels: fast (300s), thorough (600s), exhaustive (900s)
15. Trend Analyzer: Detects emerging topics via query modifiers (trending:, latest:, current:); Google Trends + Reddit API integration (when available); auto-pruning of stale topics

C. Infrastructure Agents (6)

16. Redis Cache Agent: Multi-level caching: Redis (1h TTL preferred) -> File (backend_cache.json, 24h) -> Memory fallback; automatic TTL expiration; cache invalidation
17. Model Router: Dynamic model selection per task: phi3:mini for planning (30s timeout), qwen2.5:3b for synthesis (120s timeout); per-stage timeouts in _model_router.py
18. Session Manager: Persists session state every 30s to Redis; enables true background execution via Celery; checkpoint/restore support for interruptible research
19. Monitoring Agent: Prometheus metrics (REQUESTS_TOTAL, LATENCY_HISTOGRAM); LangSmith tracing; per-node timing; error rate alerts; /api/v1/agent-status endpoint
20. Scheduler Agent: POST /api/v1/research/schedule endpoint; supports daily/weekly/monthly frequencies; integrates with Celery Beat; schedule_id generation
21. Auto-Continue Agent: Checks gap_iteration < max_gaps; if gaps exist, triggers research_orchestrator node with new queries based on missing_topics; up to 3 retries